Nicola Wengiel

Professorship for Mathematical Data Analytics
Department of Mathematics
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Education

May 2023–present	**PhD in Mathematics (in progress)** FAU Erlangen-Nürnberg. Supervisor: Prof. Dr. Timm Oertel.
October 2020–April 2023	**Master of Science (M.Sc.) in Mathematics** Faculty of Sciences, FAU Erlangen-Nürnberg. Minor: Computer Science. Thesis: *On Mixed-Integer Programs with a Disjunctive Box Constraint.* Supervisor: Prof. Dr. Frauke Liers.
October 2017–March 2021	**Bachelor of Science (B.Sc.) in Mathematics** Faculty of Sciences, FAU Erlangen-Nürnberg. Minor: Computer Science. Thesis: *On Lifted Submodular Inequalities for Capacitated Lot-Sizing.* Supervisors: Prof. Dr. Alexander Martin and Dr. Dieter Weninger.

Research Interests

Online discrepancy and prefix balancing, geometric discrepancy (especially star discrepancy), Davenport constants and zero-sum problems, and integer geometry and optimization.

Research Experience

May 2023–present	**Research Associate (Doctoral Researcher)** Department of Mathematics, FAU Erlangen-Nürnberg. Doctoral research in discrepancy theory, online balancing, and Davenport constants, using probabilistic, random matrix, and integer-geometric methods.

Publications

Dmitriy Kunisky, Timm Oertel, Nicola Wengiel, and Peiyuan Zhang (Oct. 12, 2026). "Asymptotic Bounds and Online Algorithms for Average-Case Matrix Discrepancy". In: *Random Structures & Algorithms* 69.3, e70102. DOI: 10.1002/rsa.70102. arXiv: 2410.23915 [math.PR].

Preprints

Stefan Kuhlmann, Timm Oertel, and Nicola Wengiel (Nov. 10, 2026). *Davenport Constants of High-Dimensional Integer Sets.* arXiv: 2611.14140 [math.CO].

Nicola Wengiel (Oct. 18, 2026). *Online Discrepancy for Sub-Exponential Product Distributions via Regularization.* Version 2. arXiv: 2608.10040v2 [cs.DS].

Nicola Wengiel (Oct. 25, 2026). *Online Discrepancy of Sparse Random Vectors over Long Horizons.* arXiv: 2610.34410 [cs.DS].

Nicola Wengiel (Nov. 4, 2026). *Resolving Tusnády's Problem.* arXiv: 2611.06998 [math.CA].

Additional Information

Peer review: One manuscript review for *Discrete Optimization* (2025).
Languages: German (native); English (fluent).
Technical skills: Python; LaTeX.

Teaching Experience

Teaching at bachelor's and master's levels in German and English.
Courses included: Linear and Combinatorial Optimization; Linear and Nonlinear Systems; Discrete Optimization; Mathematics for Economists; Geometry; Mathematics for Data Science and Physics; Mathematics for Engineers.

May 2023–present	**Research Associate (Doctoral Researcher)** FAU Erlangen-Nürnberg. Occasional substitute lectures; leading tutorials; preparation of examinations; teaching-related administrative duties.
October 2019–April 2023	**Student Assistant** FAU Erlangen-Nürnberg. Appointments within Collaborative Research Centre SFB/TRR154 (gas networks) and the Chair of Applied Mathematics (Mixed-Integer Linear and Nonlinear Optimization). Teaching support for Dr. Dieter Weninger within Prof. Dr. Alexander Martin's group. Literature review for teaching materials; preparation of course notes and exercise problems; leading tutorials.

Conferences and Academic Activities

6–10 July 2026	**31st British Combinatorial Conference (BCC)** Cardiff University, Cardiff, United Kingdom. Invited talk, mini-symposium Combinatorial Optimisation (7 July): *Asymptotic Bounds and Online Algorithms for Average-Case Matrix Discrepancy.*
8 May 2026	**Friday@Noon – Bavarian Workshop on Optimization** TUM Campus Straubing, Straubing, Germany. Contributed talk: *Online Discrepancy Minimization for Sub-Gaussian Inputs via Regularization and Restriction.*
18–20 March 2026	**16th Workshop on Stochastic Models, Statistics and Their Applications (SMSA)** University of Würzburg, Würzburg, Germany. Contributed talk (18 March): *Asymptotic Bounds and Online Algorithms for Average-Case Matrix Discrepancy.*
23 May 2025	**Friday@Noon – Bavarian Optimization Workshop** University of Technology Nuremberg (UTN), Nuremberg, Germany. Contributed talk: *Asymptotic Discrepancy of Gaussian Orthogonal Ensemble Matrices.*